

Hardwired and Micro-programmed Control units

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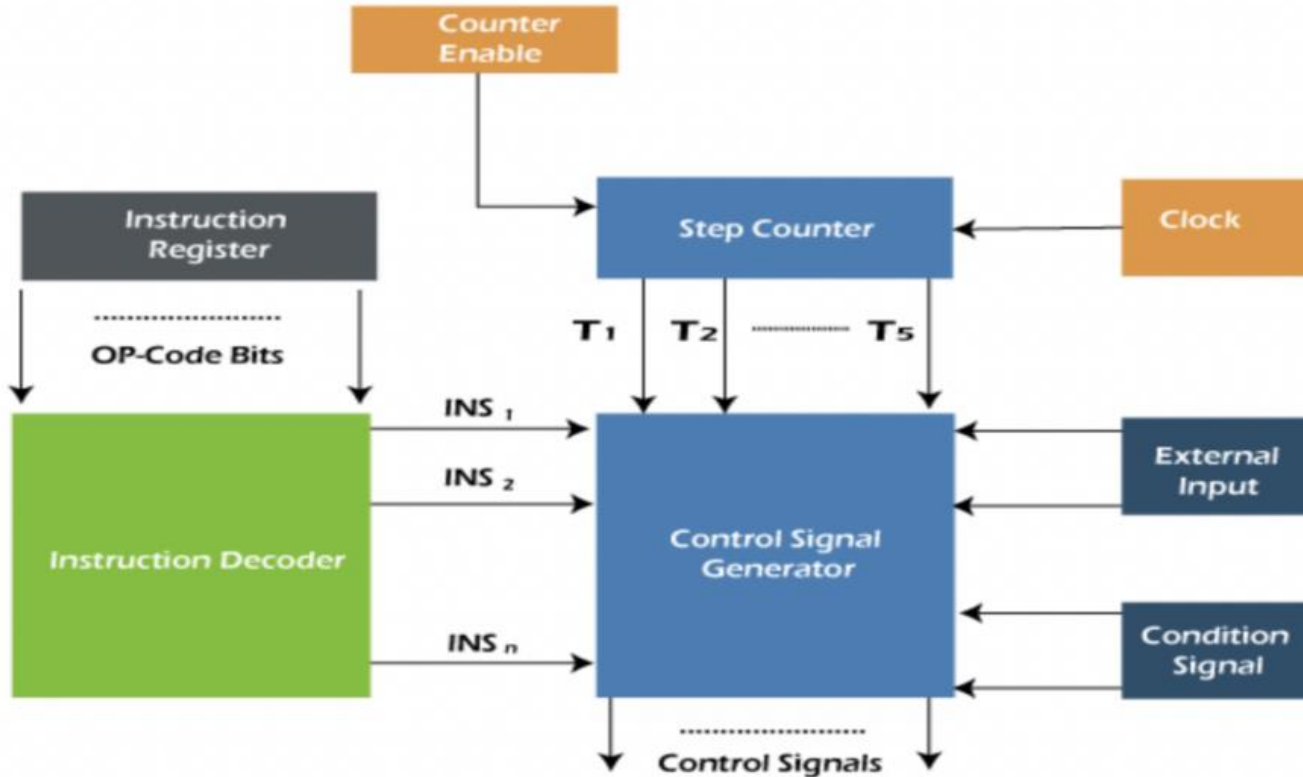
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- To execute an instruction, there are two types of control units Hardwired Control unit and Micro-programmed control unit.
 - Hardwired control units are generally faster than microprogrammed designs.
 - A microprogrammed control unit is a relatively simple logic circuit that is capable of
 - sequencing through microinstructions
 - generating control signals to execute each microinstruction.

Hardwired control unit

- The setting of the control signals depends on:

 - Contents of the step counter
 - Contents of the instruction register
 - The result of a computation or a comparison operation
 - External input signals, such as interrupt requests
- Hardwired Control Unit
 - It is a method of generating control signals with the help of Finite State Machines (FSM). It's made in the form of a sequential logic circuit by physically connecting components such as flip-flops, gates, and drums that result in the finished circuit. As a result, it's known as a hardwired controller.

Hard-wired control unit



- **Instruction register** is a type of processor register used to contain an instruction that is currently in execution. It generates the OP-code bits respective of the operation as well as the addressing mode of operands.
- The **instruction decoder** decodes the opcode. Now on the basis of the addressing mode of instruction and operation which exists in the instruction register, the instruction decoder sets the corresponding Instruction signal INS_i to 1.
- **Step Counter** - specifies the current step of instruction execution. It contains the signals from $T_1, \dots, T_5$. Now on the basis of the step which contains the instruction, one of the signals of a step counter will be set from T_1 to T_5 to 1.
- **Clock** - The one-clock cycle of the clock will be completed for each step. For example, suppose that if the step counter sets T_3 to 1, then after completing one clock cycle, the step counter will set T_4 to 1.
- **Counter Enable** will "disable" the Step Counter so that it will stop till current step of execution is complete then increment to the next step signal.
- **Condition Signals** - There are various conditions in which the signals are generated with the help of control signals that can be less than, greater than, less than equal, greater than equal, and many more.
- The **external input** is the last one. It is used to tell the Control Signal Generator about the interrupts, which will affect the execution of an instruction.

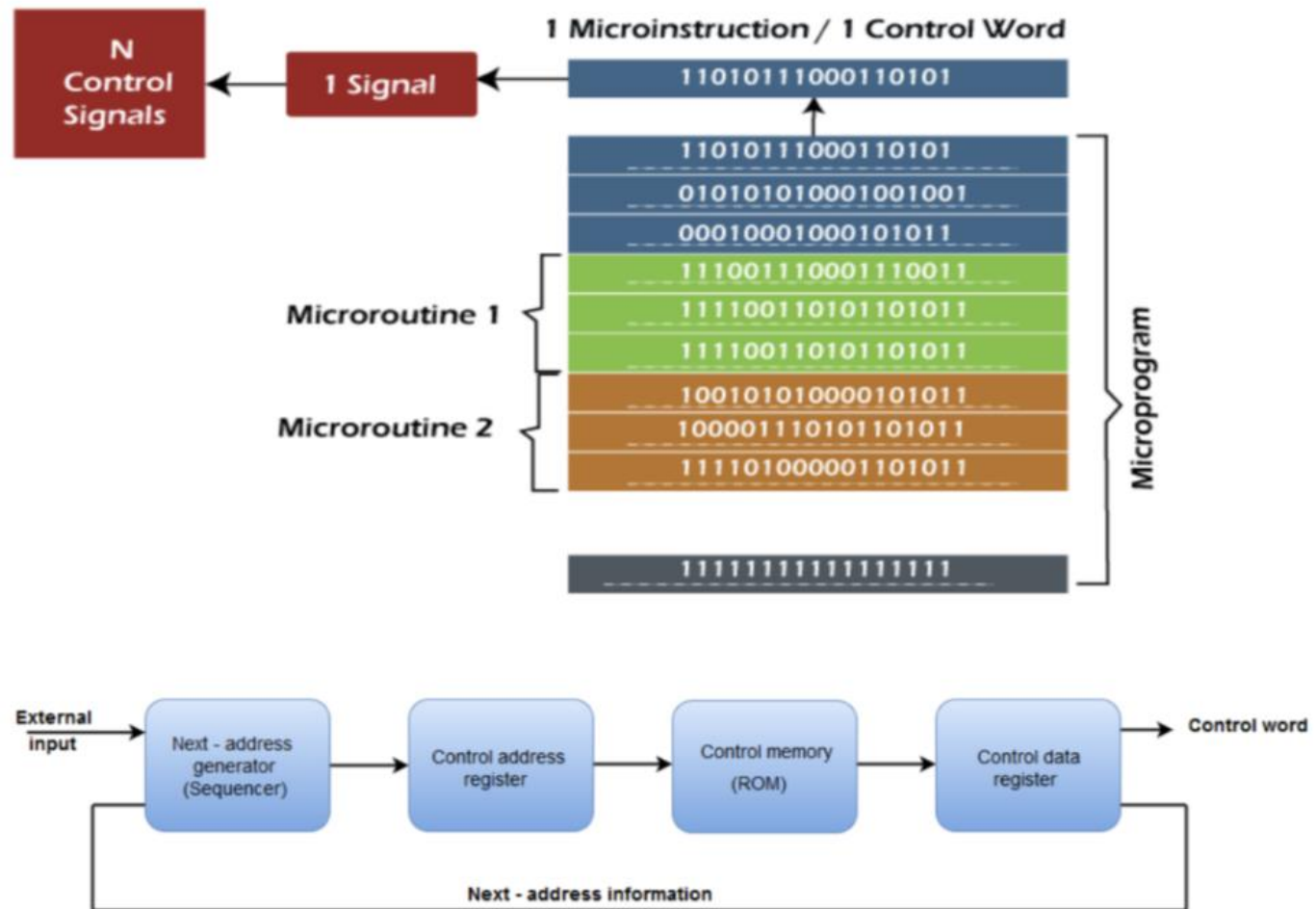
Hard-wired control unit

- Hardwired control unit is faster when compared to microprogrammed control unit as the required control signals are generated with the help of hardware's.
- Used in computer that makes use of Reduced Instruction Set Computers(RISC)

Microprogrammed control unit

- A control unit whose binary control values are saved as words in memory is called a microprogrammed control unit.
- **Control word:** A control word is a word whose individual bits represents various control signals.
- **Micro-routine:** A sequence of control words corresponding to the control sequence of a machine instruction constitute the micro routine for that instruction.
- **Micro-instruction:** Individual control words in this micro-routine is referred to as micro-instruction.
- **Micro-program:** A sequence of micro-instructions are called micro-program which is stored in a RAM or ROM called a control memory.
- **Control store:** the micro-routines for all instructions in the instruction set of the computer are stored in a special memory called the control store.

Micro-programmed control unit



Microprogrammed unit

- Control memory address register specifies the address of the micro-instruction.
- The control memory is assumed to be a ROM within which all control information is permanently stored.
- The control register holds the microinstruction fetched from memory.
- The micro-instruction contains a control word that specifies one or more micro-operations for the data processor.
- While the micro-operations are being executed, the next address is computed in the next address generator circuit and then transferred into the control address register to read the next microinstruction.
- The next address generator is often referred to as micro programmed sequencer as it determines the address sequence that is read from control memory.

Micro-programmed control unit

- This is slower than the hardwired CU, as micro instructions are used for generating signals here.
- Used in computer that makes use of Complex Instruction Set Computers(CISC)

Hardwired Control Unit	Microprogrammed Control Unit
Hardwired control unit generates the control signals needed for the processor using logic circuits	Microprogrammed control unit generates the control signals with the help of micro instructions stored in control memory
Hardwired control unit is faster when compared to microprogrammed control unit as the required control signals are generated with the help of hardwares	This is slower than the other as micro instructions are used for generating signals here
Difficult to modify as the control signals that need to be generated are hard wired	Easy to modify as the modification need to be done only at the instruction level
More costlier as everything has to be realized in terms of logic gates	Less costlier than hardwired control as only micro instructions are used for generating control signals
It cannot handle complex instructions as the circuit design for it becomes complex	It can handle complex instructions
Only limited number of instructions are used due to the hardware implementation	Control signals for many instructions can be generated
Used in computer that makes use of Reduced Instruction Set Computers(RISC)	Used in computer that makes use of Complex Instruction Set Computers(CISC)